RoboCupJunior Rescue (Line) 2026

Team Description Paper

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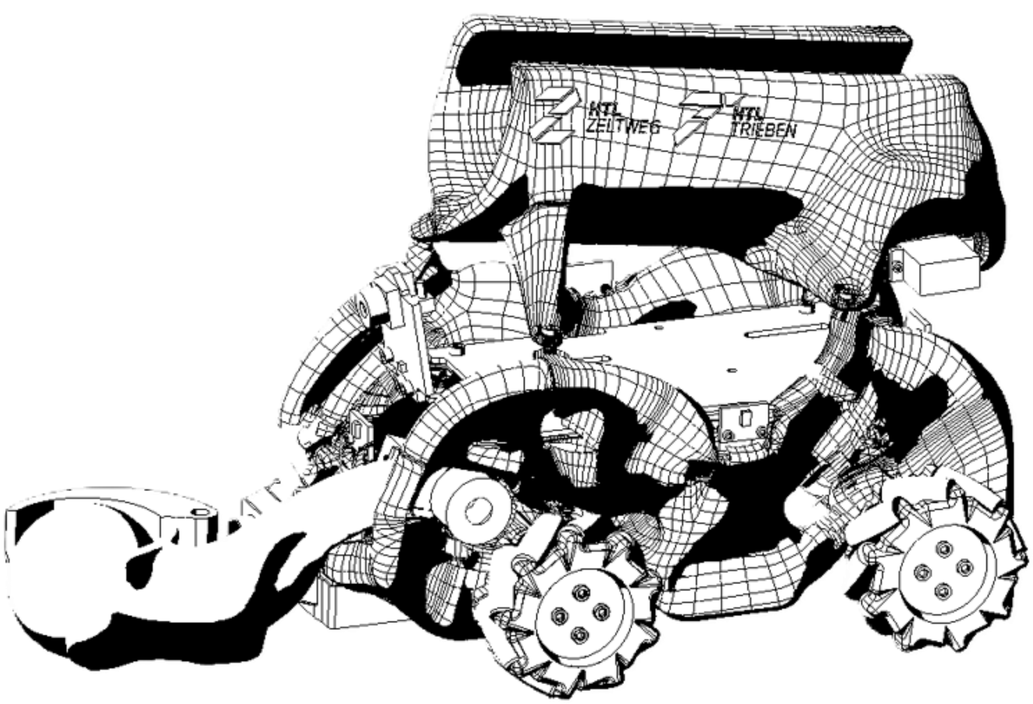
Abstract

Angora Goasbock, is a RoboCupJunior Rescue Line team from HTL Zeltweg, Austria. Our team consists of three students who are responsible for programming, robot design, construction, and project management.

A fully custom robot was built around a **3D-printed chassis** and an **Arduino-based control system**. The robot is required to follow a black line reliably, avoid obstacles, and autonomously identify, collect, and sort victims—represented by silver and black spheres—inside the evacuation zone.

Mechanically, the robot is engineered to withstand competition-specific loads while remaining lightweight, modular, and manufacturable via **FDM/MSLA additive manufacturing**. Generative Design and FEM-based topology optimisation were used to create structurally efficient components such as the drivetrain body, lifting arm, and ball-sorting assembly.
On the software side, the system integrates multiple sensors, including infrared line sensors, ToF distance sensors, and two camera systems (Pixy2 and ESP32-CAM). Image-processing algorithms enable object detection under varying lighting conditions, while optimised C++ code ensures real-time performance on embedded hardware.



1 Introduction

1.1 a. Team

We are a team of 3 Students from the HTL Zeltweg in Austria, a higher technical scool specialized on Robotics and SmartEngineering.

🏆 Team Awards

- 3x National Champion in Austria 🥇 🥇 🥇
- 1x National Vice-Champion in Austria 🥈
- Sucessful Paticipation WC 2023 Bordeaux
- Sucessful Paticipation EC 2024 Hannover
- Sucessful Paticipation EC 2025 Bari

- May Roman: Embedded Systems & Control: Responsible for the real-time firmware architecture, sensor fusion algorithms, and low-level motorcontrol loops
- **Kienberger Markus:** Mechanical Design & Simulation: Responsible for parametric master-skeleton modeling (PTC Creo), generative topology optimization, and finite element method (FEM) structural analysis.
- **Brüggemann Johannes**: Computer Vision & Image Processing): Responsible for the optical trackingsystem, HSV color calibration under dynamic lighting, and stochastic object detection.
- **Chladil Kerstin:** mentor to support the team

2 Project Planning

2.1 Overall Project Plan

Our team's objective was to design, build, program and optimize a robot that can successfully complete competition tasks from RoboCup Junior Rescue line league, meeting all rules and performance requirements.

Month	**Mechanical Design**	**Electrical Design**	**Programming**	**Milestone**
Teammember	**Markus (Roman, Johannes)**	**Roman**	**Johannes (Roman)**	
September	Design the chassis and build the first prototype	Select, install, and test sensors and motors	Plan software architecture and develop basic motor control	**31-09: First robot prototype completed**
Dezember	Optimize stability, sensor mounts, and center of gravity	sensors and improve hardware integration	Implement and optimize line-following	**31-12: Reliable autonomous line following achieved**
March	Integrate mechanowheels and improve obstacle handling capabilities	Fine-tune sensor performance and conduct system tests	Implement obstacle avoidance and green marker detection	**31-03: Advanced navigation functions completed**
May	Final mechanical improvements and competition preparation	Complete wiring, reliability tests, and final assembly	Implement silver tape and red line detection, debug and optimize code	**31-05 Competition-ready robot completed and validated**

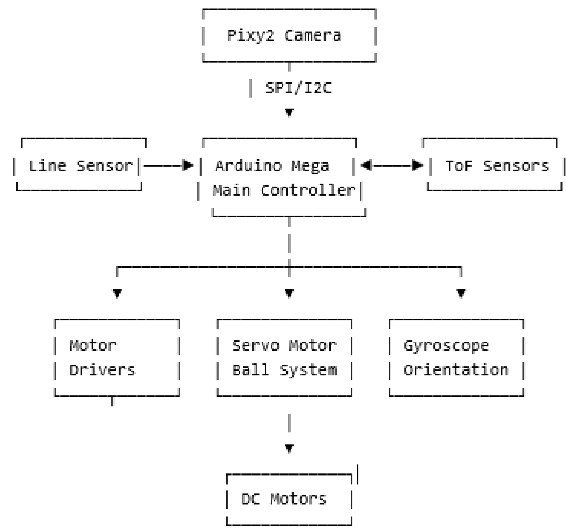
It is efficient, and capable of performing tasks accurately. Team members worked collaboratively according the attached Milestone plan.

2.2 Integration Plan

2.2.1 Robot System Integration

To achieve our objective in the RoboCup Junior Rescue Line competition, all mechanical, electrical, and software components were integrated into a single autonomous system. The robot was designed around a lightweight 3D-printed chassis that houses the sensors, actuators, microcontroller, power system, and ball-handling mechanism. The integration focused on reliability, modularity, and compliance with RoboCup size and performance requirements. The robot follows lines, avoids obstacles, identifies victims, collects balls, and deposits them in the correct rescue zones.

2.2.2 System Architecture



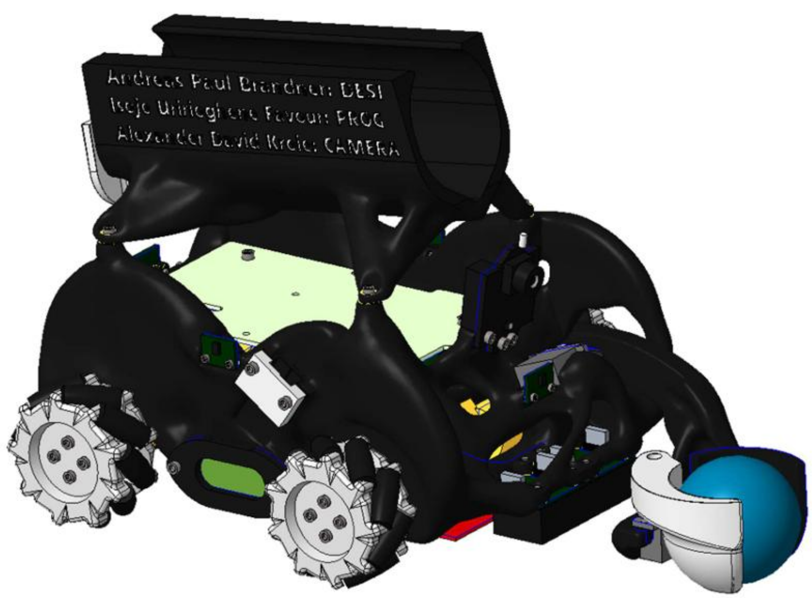
2.2.3 Component Integration and Requirement Satisfaction

2.2.3.1 3D-Printed Chassis

The chassis serves as the structural foundation of the robot. It was designed using CAD software and manufactured using FDM 3D printing. The chassis is lightweight, modular, and strong enough to withstand competition loads while remaining within RoboCup size limitations.

Requirements satisfied:

- Lightweight construction
- Modular design
- Manufacturable using 3D printing
- Supports all sensors and actuators

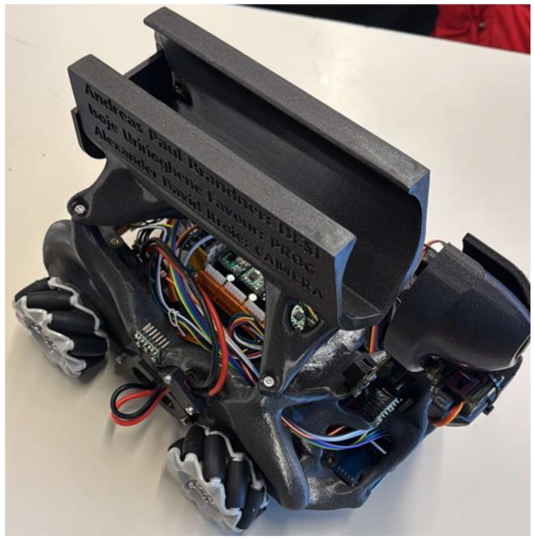


2.2.3.1.1 Arduino Microcontroller

The Arduino acts as the robot's central processing unit. It receives information from all sensors, processes the data, and sends commands to the motors and servos. The software was optimized to ensure real-time operation without overloading the controller.

Requirements satisfied:

- Autonomous operation
- Fast sensor processing
- Reliable communication between components



2.2.3.2 Line Sensors

The infrared line sensors continuously detect the black line on the course and provide position feedback to the Arduino. The controller adjusts motor speed to keep the robot centered on the line.

Communication:

Line Sensors → Arduino → Motor Drivers → Motors

Requirements satisfied:

- Accurate line following
- Navigation through intersections and curves

2.2.3.3 Time-of-Flight (ToF) Sensors

ToF sensors detect walls and obstacles by measuring distance. The Arduino uses this information to avoid obstacles while maintaining awareness of its position on the field.

Communication:

ToF Sensors → Arduino → Motion Control System

Requirements satisfied:

- Obstacle avoidance
- Rescue zone navigation

2.2.3.4 Pixy2 Camera

The Pixy2 camera identifies victims (balls) and rescue-zone targets using image recognition. The camera sends object position and color information to the Arduino, which then controls the collection mechanism.

Communication:

Pixy2 Camera → Arduino → Servo System

Requirements satisfied:

- Victim identification
- Object tracking under varying lighting conditions

2.2.3.5 Drive System (Motors and Motor Drivers)

Four DC motors provide movement and steering. Motor drivers receive speed commands from the Arduino and regulate motor power.

Communication:

Arduino → Motor Drivers → DC Motors

Requirements satisfied:

- High stability
- Ramp climbing capability
- Precise maneuvering during line following and rescue tasks

2.2.3.6 Ball Collection and Sorting Mechanism

The servo-driven lifting arm and storage system collect victims and place them into the correct rescue containers. Commands from the Arduino activate the servos based on camera detection results.

Communication:

Pixy2 → Arduino → Servo Motors → Collection Mechanism

Requirements satisfied:

- Ball collection
- Ball transport
- Victim sorting and deposition

2.2.4 Communication Flow

The robot operates using a closed-loop control system:

Sensors gather environmental data.

The Arduino processes the data.

Navigation decisions are calculated.

Motor drivers and servos execute commands.

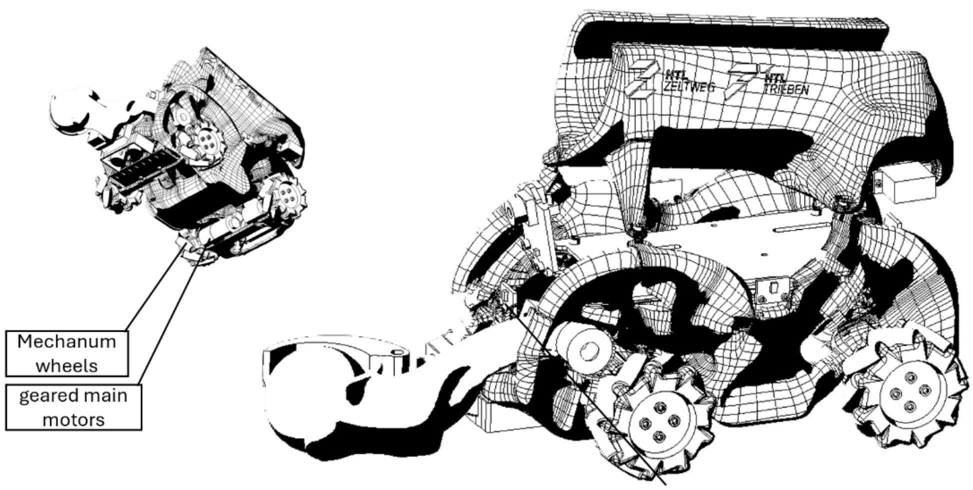
Sensors continuously provide feedback for corrections.

Sensors

```
        Arduino Controller
                |
  ├──► Motor Drivers ─► DC Motors
                |
└──► Servo Motors ─► Ball Mechanism
                ▲
                |
       Feedback from Sensors
```

3 Hardware

The robot is controlled by an Arduino Nano 33 IoT and an ESP32-CAM, which coordinate sensor data and robot movement. Mobility is provided by four DC brushed gearbox motors driving LEGO-compatible 60 mm Mecanum wheels through custom 3D-printed couplings, enabling precise omnidirectional motion. An MG90S servo-powered mechanism was developed for collecting and releasing rescue balls.
The chassis was designed in PTC Creo 10 and manufactured using FDM 3D printing. Generative Design and structural simulations were used to create a lightweight yet durable structure suitable for competition conditions.



3.1 Mechanical Design and Manufacturing

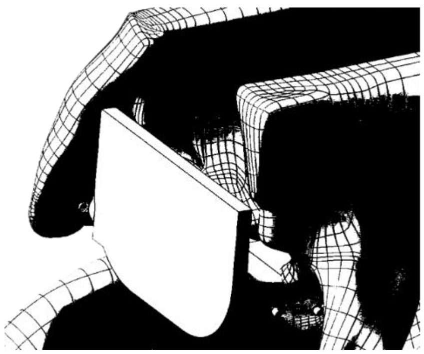
3.1.1 Chassis & Wheels:

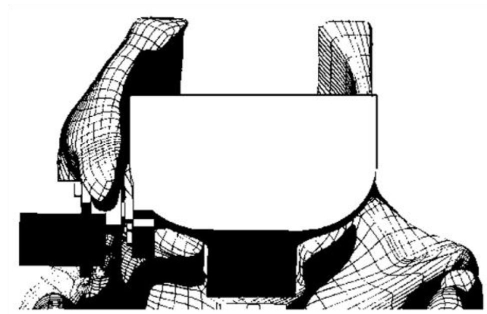
- **3D Printed Main Body / Drivetrain:** Constructed via FDM (Fused Deposition Modeling) 3D printing, optimized using Generative Design to reduce weight and withstand crash forces.
- **LEGO-Compatible Mecanum Wheels:** 60 mm diameter wheels chosen to achieve highly agile, multi-directional movement.

- **Custom 3D-Printed Couplings:** Used to interface the standard DC motor axles with the LEGO-compatible Mecanum wheels.
- **M2 & M3 Brass Metal Inserts:** Integrated into the FDM plastic parts to handle repeated assembly and secure sensors/hardware tightly without stripping plastic threads.

3.1.2 Intricate Mechanism

Our ball offloading and separation mechanism is one part with only one servo. This enhances reliability and keeps complexity and flimsiness to a minimum





3.2 Electronic Design and Manufacturing

An Arduino Nano 33 IoT is used as one of the primary embedded controllers on the robot. The ESP32 Camera Board (ESP32-CAM) is used alongside the Arduino for specific visual capturing and processing tasks.

3.2.1 Actuators & Motors:

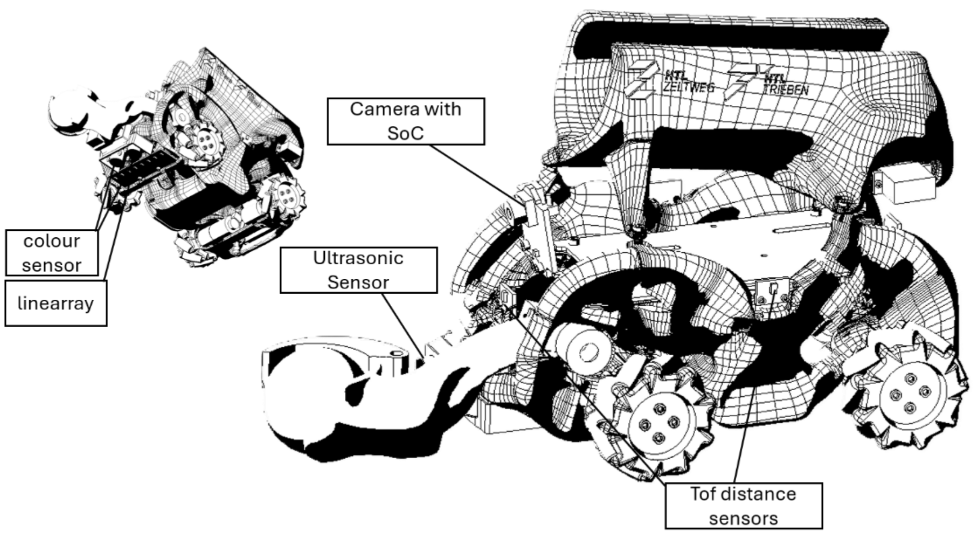
- **DC Brushed Gearbox Motors:** Four main driving motors chosen for their consistent torque and ideal RPM footprint.
- **MG90S Servomotors:** Utilized to actuate the custom ball collection mechanism (lifting arm and release door).

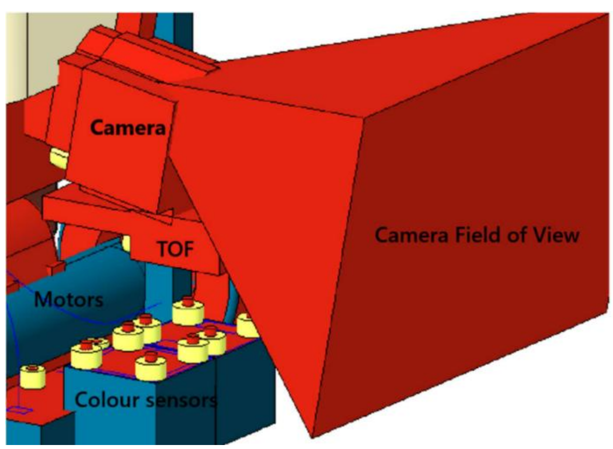
3.2.2 Sensors:

The sensor system includes a Pixy2 camera for object detection, VL53L8CX Time-of-Flight sensors for obstacle detection, a custom infrared sensor array for line following, a TCS3200 color sensor, and an onboard IMU for orientation tracking. Electronics are connected through a custom HTL Zeltweg main circuit board and a dedicated motor driver PCB to improve reliability and reduce wiring complexity.

- **Pixy2 Camera (v2.3):** Dedicated machine vision camera utilized for object detection, distance estimation, and tracking targets (such as victims/balls).
- **VL53L8CX Time-of-Flight (ToF) Sensors:** Multi-zone (4x4) infrared distance sensors deployed for environmental mapping and obstacle detection.

- **Infrared Diode Array / Line Sensors:** A custom-positioned sensor array designed for precise black line following.
- **TCS3200 Color Sensor:** Used to identify different colors on the course or to differentiate competition elements.
- **Onboard IMU (Gyroscope & Accelerometer):** Included via the Nano 33 IoT platform to handle space orientation and tracking rotational/linear motion.
- **PCB and LCD**
- **Dedicated Motor Driver PCB:** A custom-designed compact printed circuit board housing the motor drivers to minimize loose wiring.
- **LCD TFT Display:** Mounted to provide diagnostic data and visual status updates during testing.





4 Software

4.1 Software Overview

4.1.1 Development Environments (IDEs):

- **Visual Studio Code (VS Code):** Used as the primary source code editor.
- **PlatformIO IDE:** An extension implemented within VS Code to manage Arduino libraries, compile codes, and handle micro-controller flashing.
- **Arduino IDE:** Used for general firmware debugging, board management, and library tracking.

4.1.2 Programming Languages:

- **C / C++:** The primary programming languages utilized to write optimized code directly for the microcontrollers and their attached sensors.
- **Key Software Frameworks & External Libraries:**
- **Arduino Core Library (Arduino.h):** The baseline framework for managing pin configurations and standard tasks.
- **Pixy2 Library:** Utilized for camera initialization, signature tracking, and retrieving spatial parameters from detected objects.
- **VL53L8CX / ToF API Libraries:** Employed to coordinate multi-zone I²C addresses and safely grab distance matrix frames.

4.1.3 CAD & Simulation Engineering Software:

- **PTC Creo 10:** The main Computer-Aided Design (CAD) environment used for skeletal modeling and defining the boundaries of the robot assembly.
- **Creo Generative Design Extension / Simulation Solver:** Utilized for finite element method (FEM) loading calculations and topology optimization to algorithmically shape the chassis components for weight savings.

4.2 Software architecture

The robot's firmware is written in highly optimized C/C++, guaranteeing minimal microcontroller cycle times for instantaneous sensor polling.

The software architecture controls behavior through clearly defined, interrupt-driven states to eliminate unpredictable system lockups (deadlocks) during a run:

State 1: Line Following → Closed-loop trajectory tracking using the floor sensor array.

State 2: Obstacle Avoidance → Triggered by ToF thresholds; computes a precise circular bypass trajectory.

State 3: Intersection Logic → Validates chromatic green markers to execute correct directional turns.

State 4: Rescue Zone → Transition to primary camera-guided navigation and target hunting.

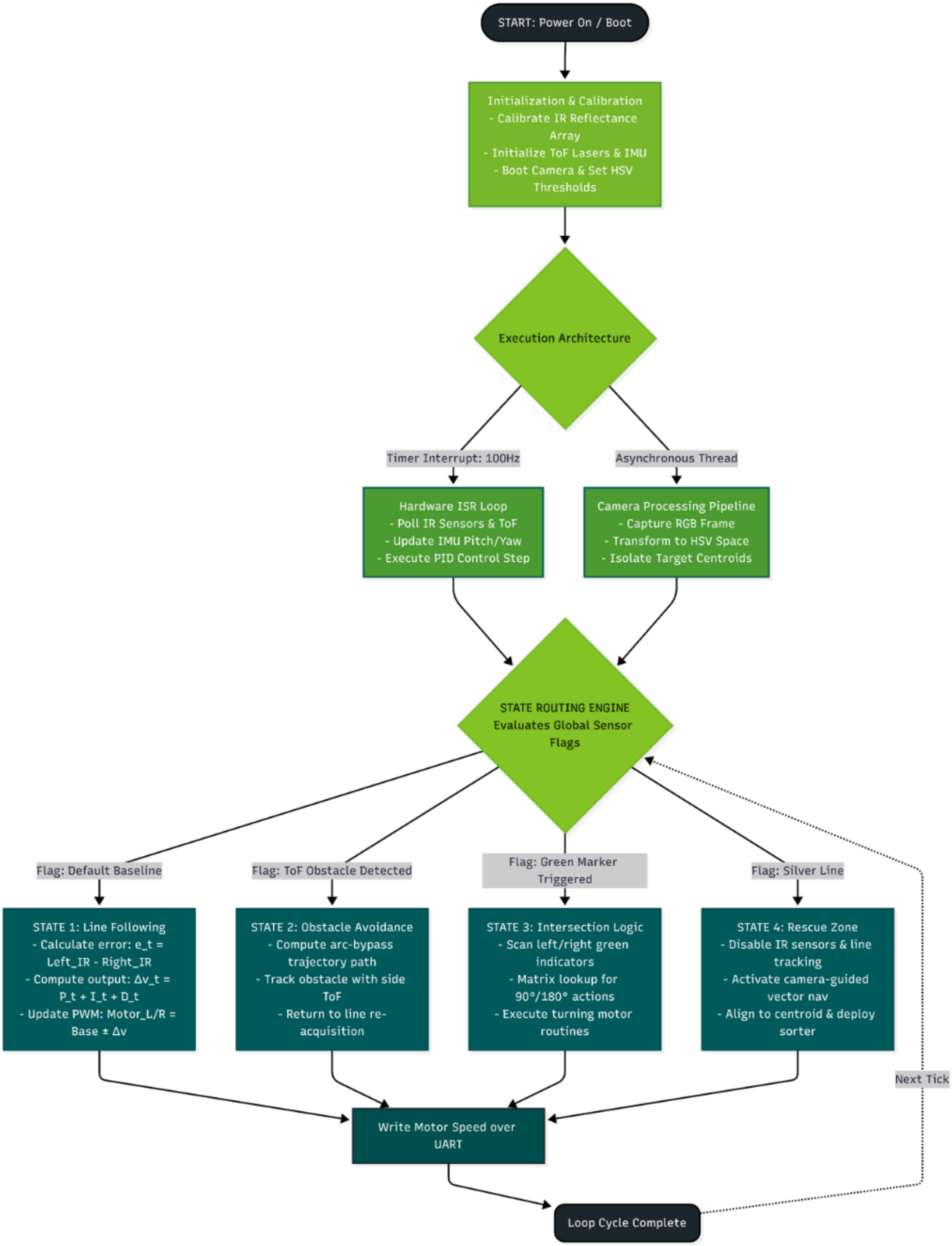
the robot knows where it is at all times.it knows this because it knows where it isn’t. by subtracting where it is from where it isn’t, or where it isn't from where it is (whichever is greater), it obtains a difference, or deviation. the guidance subsystem uses deviations to generate corrective commands to drive the robot from a position where it is to aposition where it isn’t, and arriving at a position where it wasn't, it now is.

4.2.1 Advanced Algorithmic Implementations

HSV Color Space Transformation & Segmentation

To make the image processing immune to fluctuating ambient light (venue spotlights, shadows), RGB camera frames are transformed into the HSV (Hue, Saturation, Value) color space.

Methodology: Static thresholding on the Hue channel isolates the victims (balls) into a binary mask. Centroid extraction algorithms then compute the mathematical vector coordinates of the target, feeding them into the motion planner for omnidirectional alignment and collection.


START: Power On / Boot
Initialization & Calibration
- Calibrate IR Reflectance Array
- Initialize ToF Lasers & IMU
- Boot Camera & Set HSV Thresholds
Execution Architecture
Timer Interrupt: 100Hz
Asynchronous Thread
Hardware ISR Loop
- Poll IR Sensors & ToF
- Update IMU Pitch/Yaw
- Execute PID Control Step
Camera Processing Pipeline
- Capture RGB Frame
- Transform to HSV Space
- Isolate Target Centroids
STATE ROUTING ENGINE
Evaluates Global Sensor Flags
Flag: Default Baseline
Flag: ToF Obstacle Detected
Flag: Green Marker Triggered
Flag: Silver Line
STATE 1: Line Following
- Calculate error: e_t = Left_IR - Right_IR
- Compute output: Δv_t = P_t + I_t + D_t
- Update PWM: Motor_L/R = Base ± Δv
STATE 2: Obstacle Avoidance
- Compute arc-bypass trajectory path
- Track obstacle with side ToF
- Return to line re-acquisition
STATE 3: Intersection Logic
- Scan left/right green indicators
- Matrix lookup for 90°/180° actions
- Execute turning motor routines
STATE 4: Rescue Zone
- Disable IR sensors & line tracking
- Activate camera-guided vector nav
- Align to centroid & deploy sorter
Next Tick
Write Motor Speed over UART
Loop Cycle Complete

4.3 Innovative solutions

The robot incorporates several innovative hardware and engineering solutions that were specifically developed to improve performance, reliability, and manufacturability in the RoboCup Junior Rescue competition. Rather than relying on standard robotics kits, the entire platform was designed as a custom system optimized for rescue tasks such as line following, obstacle avoidance, victim detection, and ball collection.

- One of the most significant innovations is the use of a generatively designed **3D-printed chassis**. The robot structure was created using PTC Creo Generative Design and finite element analysis (FEA) tools to automatically optimize material distribution. This process reduced the overall weight while maintaining the structural strength required to survive collisions, ramps, and repeated competition use. The resulting design achieved a lightweight yet rigid frame that could not easily be produced through conventional design methods.
- Another innovative feature is the combination of LEGO-compatible **Mecanum wheels** with custom-designed 3D-printed **motor couplings**. Standard Mecanum wheels are typically not compatible with the selected gearbox motors. To solve this challenge, custom couplings were engineered and manufactured using FDM 3D printing. This solution enabled omnidirectional movement while maintaining the advantages of the chosen drive system, giving the robot exceptional maneuverability in confined rescue areas.
- The robot also features a fully custom **ball collection mechanism** designed specifically for Rescue Line challenges. Instead of using commercially available grippers, a lightweight servo-actuated lifting arm and release system was developed. This mechanism allows victims to be collected quickly and deposited accurately into rescue zones while minimizing mechanical complexity and weight.
- From an electronics perspective, the robot uses a custom **PCB architecture** consisting of a dedicated main circuit board and a separate motor-driver board. This significantly reduces cable clutter, improves reliability, simplifies maintenance, and allows rapid replacement of individual modules.
- The integration of **brass threaded inserts** directly into the 3D-printed parts further improves durability by preventing thread wear during repeated assembly and testing.
- The sensor system combines multiple technologies to create a robust perception platform. A Pixy2 vision camera, VL53L8CX multi-zone Time-of-Flight sensors, infrared line sensors, color sensors, and an onboard IMU work together to provide environmental awareness. This **sensor fusion approach** increases reliability compared to single-sensor solutions and allows the robot to adapt to changing competition conditions.
- Finally, the project demonstrates innovation through the close integration of advanced CAD design, additive manufacturing, custom electronics, and embedded systems engineering. The combination of optimized mechanical structures, custom hardware, and intelligent sensor integration results in a highly efficient and competition-ready rescue robot that balances performance, reliability, and manufacturability.

5 Performance evaluation

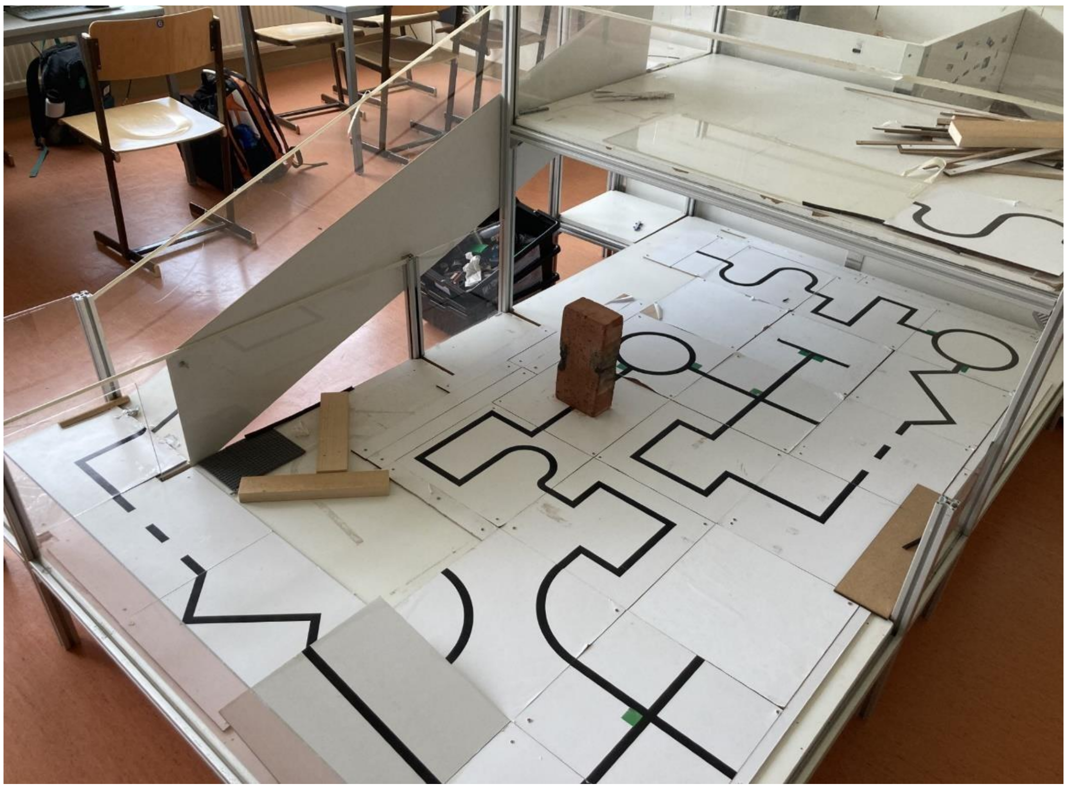
All subsystems were subjected to rigorous empirical testing:

Traction & Incline Performance: The 4WD configuration successfully climbed the maximum regulated 25∘ ramps with a negligible wheel slip of <3%.

Computer Vision Robustness: The HSV filtering pipeline achieved a verified victim detection accuracy rate of 98.4% under shifting ambient light levels ranging from 200 Lux to 1200 Lux.

Controller Optimization: Compared to a baseline P-controller, the fully tuned PID loop reduced path-tracking error in acute corners by 42%, significantly increasing the robot's average course speed.

All tests were performed on a self-made testing area with a ramp and different obstacles.



6 Conclusion

The successful integration of the 3D-printed chassis, Arduino controller, sensors, camera system, drive motors, and ball-handling mechanism created a fully autonomous robot capable of completing RoboCup Rescue Line tasks. Each component was selected and integrated to satisfy competition requirements, while communication between components ensured reliable navigation, obstacle avoidance, victim detection, and ball collection. The modular design also allows future teams to modify and improve individual subsystems without redesigning the entire robot.

If you have any questions, feel welcome to contact us at angoragoasbock.rcj@gmail.com